

SQL Queries and Subqueries

```
mysql> CREATE TABLE Employee (  
-> Employee_ID INT PRIMARY KEY,  
-> First_Name VARCHAR(50) NOT NULL,  
-> Last_Name VARCHAR(50) NOT NULL,  
-> Department VARCHAR(50) NOT NULL,  
-> Salary DECIMAL(10, 2), -- Stores monetary values (e.g., 75000.50)  
-> Hire_Date DATE, -- Stores dates as 'YYYY-MM-DD'  
-> Is_Active BOOLEAN DEFAULT TRUE  
-> );  
Query OK, 0 rows affected (0.04 sec)  
  
mysql> INSERT INTO Employee (Employee_ID, First_Name, Last_Name, Department, Salary, Hire_Date, Is_Active) VALUES  
-> (1001, 'Aditya', 'Sen', 'HR', 65000.00, '2024-03-01', TRUE),  
-> (1002, 'Kavya', 'Rao', 'Engineering', 95000.00, '2023-06-15', TRUE),  
-> (1003, 'Manish', 'Joshi', 'Marketing', 58000.00, '2025-01-10', TRUE),  
-> (1004, 'Sneha', 'Nair', 'Engineering', 105000.00, '2022-11-01', TRUE),  
-> (1005, 'Rajesh', 'Gupta', 'Finance', 72000.00, '2024-08-20', FALSE),  
-> (1006, 'Pooja', 'Mehta', 'HR', 67000.00, '2025-05-18', TRUE);  
Query OK, 6 rows affected (0.01 sec)  
Records: 6 Duplicates: 0 Warnings: 0  
  
mysql> SELECT *  
-> FROM Employee;  
  
+-----+-----+-----+-----+-----+-----+-----+  
| Employee_ID | First_Name | Last_Name | Department | Salary | Hire_Date | Is_Active |  
+-----+-----+-----+-----+-----+-----+-----+  
| 1001 | Aditya | Sen | HR | 65000.00 | 2024-03-01 | 1 |  
| 1002 | Kavya | Rao | Engineering | 95000.00 | 2023-06-15 | 1 |  
| 1003 | Manish | Joshi | Marketing | 58000.00 | 2025-01-10 | 1 |  
| 1004 | Sneha | Nair | Engineering | 105000.00 | 2022-11-01 | 1 |  
| 1005 | Rajesh | Gupta | Finance | 72000.00 | 2024-08-20 | 0 |  
| 1006 | Pooja | Mehta | HR | 67000.00 | 2025-05-18 | 1 |  
+-----+-----+-----+-----+-----+-----+-----+  
6 rows in set (0.00 sec)
```



```
mysql> SELECT First_Name, Department, Salary  
-> FROM Employee;
```

First_Name	Department	Salary
Aditya	HR	65000.00
Kavya	Engineering	95000.00
Manish	Marketing	58000.00
Sneha	Engineering	105000.00
Rajesh	Finance	72000.00
Pooja	HR	67000.00

```
6 rows in set (0.00 sec)
```

```
mysql> SELECT First_Name, Salary  
-> FROM Employee  
-> WHERE Salary > 70000.00;
```

First_Name	Salary
Kavya	95000.00
Sneha	105000.00
Rajesh	72000.00

```
3 rows in set (0.01 sec)
```

```
mysql> SELECT First_Name, Salary  
-> FROM Employee  
-> WHERE Department = 'Engineering';
```

First_Name	Salary
Kavya	95000.00
Sneha	105000.00

```
2 rows in set (0.00 sec)
```

```
mysql> SELECT First_Name, Hire_Date  
-> FROM Employee  
-> WHERE Department = 'HR'  
-> AND Hire_Date > '2024-01-01';
```

```
+-----+
```



```
mysql> SELECT First_Name, Hire_Date  
-> FROM Employee  
-> WHERE Department = 'HR'  
-> AND Hire_Date > '2024-01-01';
```

First_Name	Hire_Date
Aditya	2024-03-01
Pooja	2025-05-18

2 rows in set (0.00 sec)

```
mysql> SELECT MAX(Salary) AS Highest_Salary  
-> FROM Employee;
```

Highest_Salary
105000.00

1 row in set (0.01 sec)

```
mysql> SELECT AVG(Salary) AS Average_Salary  
-> FROM Employee;
```

Average_Salary
77000.000000

1 row in set (0.00 sec)

```
mysql> SELECT First_Name, Salary  
-> FROM Employee  
-> WHERE Salary > (  
-> SELECT AVG(Salary)  
-> FROM Employee  
-> );
```

First_Name	Salary
Kavya	95000.00
Sneha	105000.00



```
mysql> SELECT First_Name, Salary
-> FROM Employee
-> WHERE Salary = (
->     SELECT MAX(Salary)
->     FROM Employee
-> );
```

First_Name	Salary
Sneha	105000.00

1 row in set (0.00 sec)

```
mysql> SELECT First_Name, Salary
-> FROM Employee
-> WHERE Salary > (
->     SELECT Salary
->     FROM Employee
->     WHERE First_Name = 'Aditya'
-> );
```

First_Name	Salary
Kavya	95000.00
Sneha	105000.00
Rajesh	72000.00
Pooja	67000.00

4 rows in set (0.00 sec)

```
mysql> SELECT First_Name, Department
-> FROM Employee
-> WHERE Department = (
->     SELECT Department
->     FROM Employee
->     WHERE First_Name = 'Aditya'
-> );
```

First_Name	Department
Aditya	HR
Pooja	HR



Aditya	HR
Pooja	HR

2 rows in set (0.00 sec)

```
mysql> SELECT First_Name, Salary
-> FROM Employee
-> WHERE Department = 'Engineering'
-> AND Salary > (
->     SELECT AVG(Salary)
->     FROM Employee
->     WHERE Department = 'Engineering'
-> );
```

First_Name	Salary
Sneha	105000.00

1 row in set (0.00 sec)

```
mysql> SELECT First_Name, Department
-> FROM Employee
-> WHERE Department IN (
->     SELECT Department
->     FROM Employee
->     WHERE Hire_Date LIKE '2023%'
-> );
```

First_Name	Department
Kavya	Engineering
Sneha	Engineering

2 rows in set (0.01 sec)

```
mysql> SELECT First_Name, Salary
-> FROM Employee
-> WHERE Salary > ALL (
->     SELECT Salary
->     FROM Employee
->     WHERE Department = 'HR'
```

```

+-----+-----+
| First_Name | Department |
+-----+-----+
| Kavya      | Engineering |
| Sneha      | Engineering |
+-----+-----+
2 rows in set (0.01 sec)

```

```

mysql> SELECT First_Name, Salary
-> FROM Employee
-> WHERE Salary > ALL (
->     SELECT Salary
->     FROM Employee
->     WHERE Department = 'HR'
-> );

```

```

+-----+-----+
| First_Name | Salary      |
+-----+-----+
| Kavya      | 95000.00    |
| Sneha      | 105000.00   |
| Rajesh     | 72000.00    |
+-----+-----+
3 rows in set (0.00 sec)

```

```
mysql> |
```